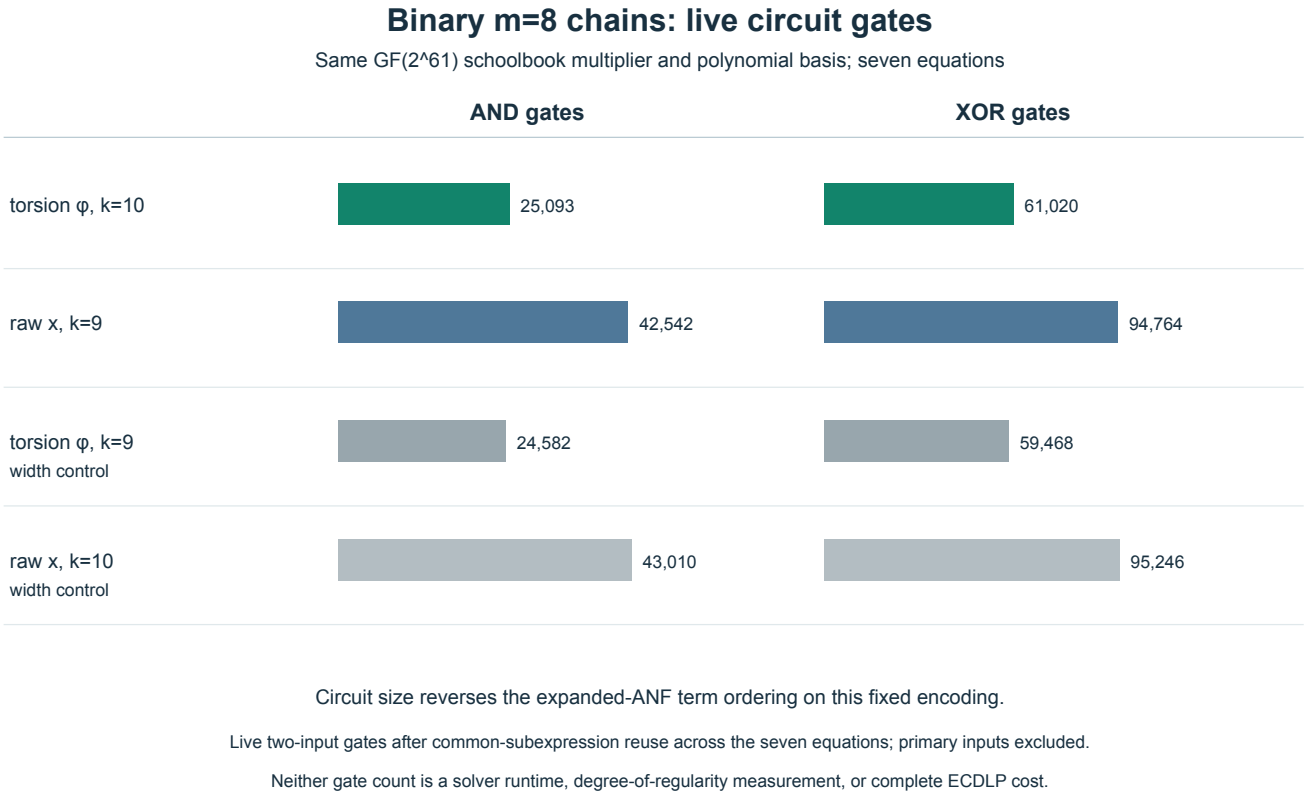
Circuit input support of two binary eight-summand chains

2026-10-17 · Structural input comparison on one GF(2^61) ordinary-binary fixture

Seven three-point equations encode one planted eight-summand raw target. The selected torsion-coordinate base has 267 projected columns and 10 external bits per coordinate; the selected raw-x base has 265 columns and 9 bits. Both use six full-field intermediates. A fixed schoolbook GF(2^61) multiplier, linear squaring/constant multiplication, two-input XOR and AND gates, and common-subexpression reuse define the circuit encoding. Counts below include only gates reachable from all 427 output bits, across the seven equations; input bits are separate.

Arm	Inputs	AND	XOR	Live gates
Torsion φ, selected	446	25,093	61,020	86,113
Raw x, selected	438	42,542	94,764	137,306
Torsion φ, width control	438	24,582	59,468	84,050
Raw x, width control	446	43,010	95,246	138,256



The circuit generator matched all seven archived field-equation values on eight seeded assignments per arm. The parent equations were independently replayed in Sage. The equal-width controls are symbolic only and have no matched realized point sets. The selected torsion form uses fewer circuit gates while its expanded ANF has more coefficient-bit terms than raw x (647,014 versus 121,546 in one middle equation). This reflects representation and shared products; it does not measure solver degree, runtime, relation yield or end-to-end cost.

Boundary. The encoding omits curve lifts, sign consistency, poles, three other torsion target channels, certificates, rank and linear algebra. A frozen solver comparison must measure these, equal realized-column controls, planted and unseen targets, full costs and matched rho. Source, exact record and links: `docs/reports/2026-10-17-binary-circuit-support.md`.